# Wormhole teleportation with a Gaussian sparse SYK instance: circuits for IBM Nighthawk

Gyalpo Dongo and Sebastian Alberdi
Ohara Labs

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**Abstract**

In response to a question from the authors of [1], we give one $N$=8 Gaussian sparse SYK instance with $K$=24 terms for the traversable-wormhole teleportation protocol, with routed circuits for IBM Nighthawk; it passes every gate of the frozen verifier. Its routed circuits cost 216/225 two-qubit gates (FIX/SYM protocols), 23% and 5% *more* than their binary Eq. 11 compiled with the same pipeline (175/215), and 30% and 45% fewer than a typical $K$=24 draw (309/410). In a FakeNighthawk noise simulation the FIX circuit still keeps 25 to 35% more teleportation signal than Eq. 11 over three routing-seed lists, and 18% more when the term order is optimized on both sides; this is our robust claim. SYM keeps 25 to 29% more in canonical term order but is at best a tie when the term order is optimized. Nothing has run on hardware.

**Problem.** Two coupled Sachdev–Ye–Kitaev (SYK) models in a thermofield double (TFD) state model a traversable wormhole: a qubit swapped into the left side at $-t_0$ reappears on the right at $t_1$ when the left-right coupling $e^{i\mu V}$ has $\mu < 0$ [2, 3, 4]; small instances have run on hardware [5], with debate on how well they scramble [6]. Sparse SYK keeps $K$ of the $\binom{N}{4} = 70$ four-Majorana terms and stays chaotic above a threshold $K$ [7, 8, 9]. Ref. [1] measures the signal $\Delta I = I_{PT}(\mu{=}-12) - I_{PT}(\mu{=}+12)$, with $I_{PT}$ the base-2 mutual information between a reference qubit $P$ and the readout qubit $T$, at $\beta = 3$ on 10 qubits, in the symmetric (SYM, $t_0 = t_1$) and fixed-injection-time (FIX, $t_0 = 1.8$) protocols, and ran a binary $K$=10 instance (their Eq. 11) on IBM hardware. They asked us for a Gaussian one ($\mathrm{Var}\, J = 1.640625/K$, $K \approx 24$) with large $\Delta I$ in both protocols and few two-qubit (2q) gates after routing on IBM Nighthawk (our bar: $\leq 0.70\times$ the typical median, FIX 216.3, SYM 287.0).

**Result.** An *instance* is one term set plus one Gaussian coupling draw. Table 1 lists the three instances that passed every gate, plus two controls; Appendix A prints the recommended instance, 19277_201. The first two share a term set. The third, 1718_714, passes by the pre-registered rule on seed list A (FIX +2.8%, SYM +0.8%) but not on held-out lists B and C (FIX −0.05%, −3.6%; SYM −1.4%, −3.1%): borderline. 19277_201 keeps +25–35% / +25–29% more noisy $\Delta I$ than Eq. 11 and 8.1 / 9.4 times a typical draw. The funnel: 32 instances on 16 term sets passed the cost, chaos, noiseless-signal and TFD-fit checks; 23 on 14 sets also passed robustness (cost on three seed lists; eight noiseless $\Delta I$ peaks $\geq 1.02\times$ Eq. 11's; $\geq 45$ of 50 draws with 1% coupling noise keep every peak above Eq. 11's); 3 on 2 sets passed the noise test. The code repository ships 16 instances: 14 of the 32 that passed (including the three in Table 1) and 2 that failed only the SYM cost bar on list B (290 and 291 against 287.0). The cheapest instance that passed (FIX 197, $K$=25) fails the noise test.

*Chaos.* 19277_201 is consistent with the $K$=24 ensemble: level-spacing ratio $r = 0.44$ / 0.45 in the two parity sectors (26th / 28th percentile; Poisson 0.386, GOE 0.53) and spectral form factor (SFF) at the 65th percentile, both within the bands; its out-of-time-order correlator (OTOC) decays faster than 91% of typical draws (decay time 2.94 against our bar 2.95). With 8 levels per sector, $r$ is weak evidence. The time-averaged SFF lies inside the $K$=24 band (5th–95th percentile of 2000 draws) in 18 of 18 time windows (155_129: 17 of 18), with log-log ramp slope 1.08 (Eq. 11: 0.53); the size-winding phase reverses after the kick for all 8 insertions in all three Hamiltonians; the two-point function's mean half-decay time is 2.08 (Eq. 11: 1.96), with late $|G|$ on $[8, 20]$ of 0.072 (0.069). A selected sparse instance can carry a signal without the chaos behind the protocol (Sec. 5 of [9]; [6]), so we also checked mirror symmetry: no exact $E \to -E$ map exists, and the near-mirror weight $m(H) = 0.217$ sits at the 98th percentile of the ensemble (farther from a mirror than 98% of draws).

*Selection.* 19277_201 is about 1 in $1.5 \times 10^{13}$ draws from a defined pool (uniform $K$=24 term sets with i.i.d. Gaussian couplings: $5 \times 10^8$ term-set draws for the 10 sets whose couplings we screened, times $7.0 \times 10^7$ coupling draws for 231 screen passes), so its exact pair-(1,2) $\Delta I$ peak (0.39 / 0.30) is 2.7 / 2.4 times the typical median (0.15 / 0.12). The "8 to 9 times typical" is at circuit level, where Trotter error, routed cost and selection compound.

*New beyond [1].* They already report Gaussian $K_c$=24 $I_{PT}$ curves (Fig. S15) and a one-step $K$=24 cost of 81 2q gates (Table S1). Our addition is one explicit Gaussian $K$=24 instance with routed circuits predicted to keep more FIX signal than Eq. 11 under the same noise model.

**Method.** Term sets: $5 \times 10^8$ uniform $K$=24 term-set draws ranked by a cheap predicted cost, 30,842 kept, the top 200 routed on FakeNighthawk. Couplings: up to $2 \times 10^7$ i.i.d. Gaussian draws per set, screened on chaos, the eight $\Delta I$ peaks and Trotter error; the couplings are selected draws, never continuously optimized. Circuits: one Lie–Trotter step (FIX) or Suzuki-2 step (SYM), the Jordan–Wigner map S2 of [1] for both protocols, canonical greedy term order, per-block pytket 2.18.5 GreedyPauliSimp and FullPeepholeOptimise, then Qiskit 2.5.2 at optimization level 3 on FakeNighthawk;

| Instance | $K$ | $t_1$ | 2q | noiseless $\Delta I$ | noisy $\Delta I$ FIX | noisy $\Delta I$ SYM | vs Eq. 11 | × typical |
|---|---|---|---|---|---|---|---|---|
| 19277_201 (set 1) | 24 | 2.6 / 2.35 | 216 / 225 | 0.33 / 0.28 | $0.116 \pm 0.005$ [0.120] | $0.096 \pm 0.0006$ [0.094] | +25–35% / +25–29% | 8.1 / 9.4 |
| 155_129 (set 1) | 24 | 2.45 / 2.15 | 216 / 225 | 0.27 / 0.25 | $0.097 \pm 0.004$ [0.102] | $0.087 \pm 0.0007$ [0.085] | +6–14% / +14–18% | 6.8 / 8.5 |
| 1718_714 (set 2)[b] | 24 | 2.55 / 2.5 | 208 / 250 | 0.25 / 0.26 | $0.092 \pm 0.002$ [0.094] | $0.075 \pm 0.0003$ [0.075] | −4 to +3% / −3 to +1% | 6.4 / 7.2 |
| Eq. 11 (binary) | 10 | 2.35 / 2.05 | 175 / 215 | 0.21 / 0.20 | $0.089 \pm 0.0009$ [0.083] | $0.074 \pm 0.001$ [0.074] | ref. | 6.3 / 7.2 |
| Typical $K$=24 ($n$=20) | 24 | – | 309 / 410 | 0.060 / 0.079[a] | 0.014 | 0.010 | −84% / −86% | 1 / 1 |

**Table 1:** Instances that pass every gate, and controls. Pairs are FIX / SYM. Sets 1 and 2 are term sets d6a76272 and 2f2d8f4f in the repository. $t_1$: readout time ($t_0 = 1.8$ for FIX, $t_0 = t_1$ for SYM). 2q: routed cz count, the lower median over 20 routing seeds, also the shipped circuit's count. Noiseless $\Delta I$: routed circuit with the fitted TFD at $t_1$, bits. Noisy $\Delta I$: FakeNighthawk gate and readout noise, median over the central 5 of 20 routing seeds (sorted by 2q, depth, seed) of seed list A; ± is half the range of the medians over seed lists A, B, C; [ ] is the shipped circuit alone. "vs Eq. 11": range over lists A, B, C, each against Eq. 11 on the same list; ratios use unrounded medians. "× typical": list A. Typical: median of 20 uniform $K$=24 draws, list A; [a]over the 17 draws with all 5 seeds; 2q is the 200-draw median. [b]borderline.

readout times, TFD fit and Trotter error in Appendix B. Noise: Aer density-matrix simulation from FakeNighthawk (qiskit-ibm-runtime 0.49.0) on the shipped circuits, two-qubit tomography of $P$ and $T$.

**Error bars and the hardware run.** Across the five central routing seeds of list A, 19277_201 ranges 0.106–0.121 (FIX) and 0.091–0.099 (SYM); every seed is above Eq. 11's median, and on held-out list B every seed beats every Eq. 11 seed. A single shipped circuit (Table 1, [ ]) moves by up to about 8% from the median. Our gate + readout noise has no idle decoherence (close to ideal dynamical decoupling, DD); adding idle $T_1/T_2$ is close to no DD. Ref. [1] ran with XpXm DD and measurement twirling, so the device should sit between the two. With idle noise, 19277_201 keeps +21% / +22% (0.078 / 0.066 vs 0.064 / 0.054), and 155_129 ties Eq. 11 on FIX (0.0642 vs 0.0644). The expected device gap to Eq. 11 is therefore about 0.01–0.03 bits. Run plan: 2 signs × 9 tomography settings × 14,900 (FIX) or 18,000 (SYM) shots, about 270k (FIX) and 320k (SYM) shots per protocol per Hamiltonian, for a standard error of 0.005 bits on $\Delta I$ (the FIX gap of 0.027 bits is then about $3.8\sigma$); interleave Eq. 11 in the same session; a typical-draw control is optional.

**Verification.** Eq. 11 gives $r = 0.537163$ / 0.478812 against the printed 0.537 / 0.479; Eq. 11's exact pair-(1,2) $\Delta I$ peaks, 0.2127 (SYM) and 0.2765 (FIX), match Fig. S1 of [1] (0.212 and 0.276, read to ±0.003). Their Table S1 transition points: Appendix B. Only verdicts of a frozen verifier (checks and bars fixed before scoring) count; seed list A searched, B certified, C held out from the search. A blind re-derivation from the shipped files alone reproduced every number needed to run the shipped circuits to $2.8 \times 10^{-7}$ (two text errors found and fixed, no number errors). We assume FakeNighthawk (revision 1) has the coupling map of the Nighthawk r2 device.

**Limits.** (i) FakeNighthawk is a simulated proxy for the device. With every FakeNighthawk error rate scaled up, 155_129 FIX falls to Eq. 11 at about 1.5 times the rates, while 19277_201 stays 16–25% above it up to 1.75 times. (ii) With term order optimized on both sides, Eq. 11 SYM routes to 134–135 2q; 19277_201 FIX keeps +18%, but SYM loses in two of three order searches (−10%; −10% / −11% on lists A / B) and ties in the third (+1% / −7%). (iii) Thin margins: 19277_201 FIX costs 216 against the bar 216.3 (list C: 217 against 219.1); its OTOC decay time is 2.94 against 2.95 and $F$ 0.965 against 0.96; the second robust instance came from a search on the set that had just passed; the final search sent 6 instances on 5 sets to the verifier and only the borderline one passed, by +0.8% on SYM, within the 8–14% per-seed spread. (iv) Eq. 11's Trotter conventions are reconstructed (Appendix B). (v) Ref. [1] printed 150 2q for Eq. 11 FIX in map S2; our pipeline gives 175 in greedy order and 226 in their printed order, a gap we cannot explain. For SYM they used map S3 and printed 282; we get 274. With S3 on both sides both robust instances still pass (Appendix B).

**Code.** `https://github.com/oharalabsai/syk-gaussian-n8`: `uv run verify` re-derives Eq. 11's known answers and every instance's 2q and noiseless numbers in about a minute; `uv run noise <instance>` gives its noisy $\Delta I$.

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**AI contribution.** The authors are responsible for all content. An AI agent harness ran the search, simulations and first drafts under human direction; every number comes from a frozen verifier or a logged job. No AI is an author.

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**Appendix A: the recommended instance 19277_201.** $H_a = \sum_{T=1}^{24} J_T\, \psi_a^i \psi_a^j \psi_a^k \psi_a^l$ with $T = (i\,j\,k\,l)$, $a = L, R$, the same convention as Eq. 11 of [1] ($\{\psi^i, \psi^j\} = \delta_{ij}$, ordered product, the factor $i^{q/2}$ absorbed in $J_T$), $J_T$ drawn i.i.d. from $\mathcal{N}(0, 1.640625/24)$.

| $T$ | $(a\,b\,c\,d)$ | $J_T$ | $T$ | $(a\,b\,c\,d)$ | $J_T$ | $T$ | $(a\,b\,c\,d)$ | $J_T$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 2 4 7 | +0.054155 | 9 | 1 4 5 8 | +0.354397 | 17 | 2 4 5 6 | +0.619194 |
| 2 | 1 2 5 7 | −0.434389 | 10 | 1 4 6 7 | +0.275300 | 18 | 2 4 6 7 | +0.008317 |
| 3 | 1 2 6 7 | +0.160388 | 11 | 2 3 4 5 | +0.032474 | 19 | 2 5 6 8 | −0.092359 |
| 4 | 1 2 6 8 | +0.333668 | 12 | 2 3 4 7 | +0.344660 | 20 | 2 5 7 8 | −0.118641 |
| 5 | 1 3 4 5 | +0.078696 | 13 | 2 3 4 8 | +0.218100 | 21 | 2 6 7 8 | +0.215055 |
| 6 | 1 3 5 6 | −0.314504 | 14 | 2 3 5 7 | +0.380139 | 22 | 3 4 5 7 | +0.330082 |
| 7 | 1 3 5 8 | −0.105945 | 15 | 2 3 6 7 | +0.336518 | 23 | 3 4 5 8 | −0.102157 |
| 8 | 1 3 6 7 | −0.019269 | 16 | 2 3 6 8 | +0.111073 | 24 | 5 6 7 8 | +0.207052 |

Trotter term order (both protocols, both evolution stages): 1 4 8 22 2 5 10 12 17 21 3 6 14 16 23 24 7 11 18 19 9 13 15 20 (the greedy commuting order of the sorted term list). Times: FIX $t_0 = 1.8$, $t_1 = 2.6$; SYM $t_0 = t_1 = 2.35$; $\beta = 3$, $\mu = \mp 12$. Shipped circuits (routing seed, 2q): FIX 1019, 216; SYM 1018, 225. The VQA angles, layouts and QASM files are in the repository.

**Appendix B: conventions and further checks.** Readout time: $t_1$ is the argmax of the exact $I_{PT}(\mu{=}{-}12)$ on $[1, 4]$ (grid 0.05); FIX keeps $t_0 = 1.8$, SYM uses $t_0 = t_1$ (values in Table 1). Scanning $t_1$ by $\pm 0.2$ on the shipped circuits, both instances beat Eq. 11 at every matched offset (worst case 1.21× FIX, 1.14× SYM, 155_129 at its $t_1$ against Eq. 11's best point), and FIX $\Delta I$ rises about 2% at +0.2 for all three, since $t_1$ maximizes $I_{PT}$, not $\Delta I$. TFD: a 6-layer, 96-angle variational circuit applied to the infinite-temperature state $|I\rangle$, best of 720 random starts; fidelity $F = 0.965$ for 19277_201 (Eq. 11: 0.98, same procedure; our bar $F \geq 0.96$). With FIX $t_0$ set to the SYM argmax instead (2.35 / 2.15 / 2.05 for 19277_201 / 155_129 / Eq. 11), the exact FIX pair-(1,2) peak drops 3–4% for all three (19277_201 0.379 vs 0.395, Eq. 11 0.267 vs 0.276) and the ratio to Eq. 11 is unchanged (1.42 vs 1.43). Trotter: the $t_0$ stage evolves $H_L$ only, and backward evolution reuses the forward term order. The Trotter error $\eta = |\max \Delta I_{\text{exact}} - \max \Delta I_{\text{Trotter}}| / \max \Delta I_{\text{exact}}$ on $[1, 4]$ with the fitted TFD is 0.094 (FIX) / 0.0017 (SYM) for 19277_201, against 0.162 / 0.012 for Eq. 11 (0.168 / 0.009 with the exact TFD). Eq. 11's noiseless Trotter peaks (FIX / SYM) are 0.2301 / 0.2147 under a term order and an $H_L$-only first stage that we reconstructed; the literal reading of [1] gives 0.2246 / 0.2130. Both sides use the reconstruction. Map S3 on both sides: 19277_201 +30% / +28%, 155_129 +19% / +15% over Eq. 11 (FakeNighthawk gate + readout, list A). Transition points: with 10,000 realizations per $K$ by their SI Sec. S2 method, the Gaussian points of their Table S1 come out at $K_1 = 22$ [21, 24] against 23 and $K_2 = 20$ against 23 (weakly constrained, [18, 69]), both Gaussian points inside our 95% intervals; binary $K_1/K_2 = 6$ [6, 7] / 9 [9, 12] against 6/8.